

TissueGraph

Hierarchical sets and operators for differentiable tissue

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Abstract

Living tissues couple biochemical signalling, cellular interaction, mechanical force, and collective dynamics across scales. The frameworks built so far (CELLGRAPH, NEURALGRAPH, MPMGRAPH, METABOLISMGRAPH) each implement *one* graph type. A tissue needs them coexisting. We define the missing primitive: a **hierarchical graph container** stated in the language of **sets** and **operators**. A tissue is a stack of sets linked by containment; dynamics are operators — GNNs implementing ODE vector fields and Laplacians — dispatched by relation. Every existing framework becomes a set of registered operators plus a schedule.

1 Two entities: sets and fields

A **set** S_k is a collection of like nodes at scale k (molecules, particles, cells, populations). The hierarchy is a sequence of **partitions**

$$\pi_k : S_k \longrightarrow S_{k+1},$$

so that a cell *is* the fibre $\pi_1^{-1}(c)$ of its particles. A **field** $f : \Omega \rightarrow \mathbb{R}^c$ is a continuum on its own grid, bound to one level. Fields do not nest; they form a flat set, each tagged with the level it couples to (Fig. 1).

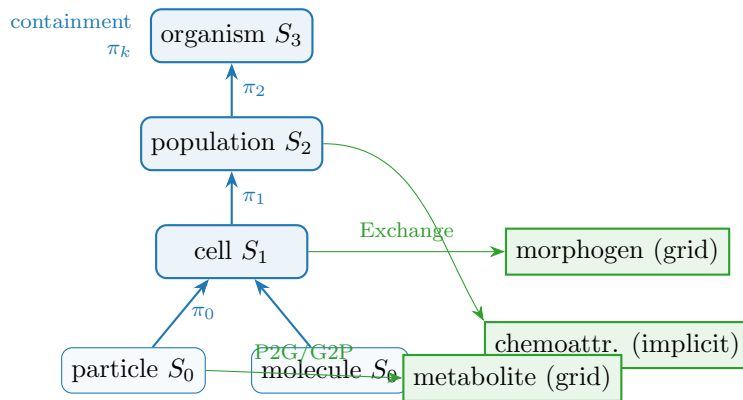


Figure 1: The container. **Blue**: sets linked by containment π_k (the tree). **Green**: fields, flat, each on its own grid, each bound to one level by an *Exchange* relation.

2 Four operators

A GNN is an **operator** $\mathcal{O}$ returning a time-derivative contribution, dispatched by the relation it acts on. There are exactly four kinds (Fig. 2):

Lateral	$\mathcal{O}_E, E \subseteq S_k \times S_k$	(ODE interaction + graph Laplacian)
Aggregate	$\uparrow \sum_{\pi} : S_k \rightarrow S_{k+1}$	(reduction over fibres of π)
Broadcast	$\downarrow \pi^* : S_{k+1} \rightarrow S_k$	(lift along π)
Exchange	$\mathcal{S}, \mathcal{G} : S_k \leftrightarrow F$	(scatter / gather; bipartite)

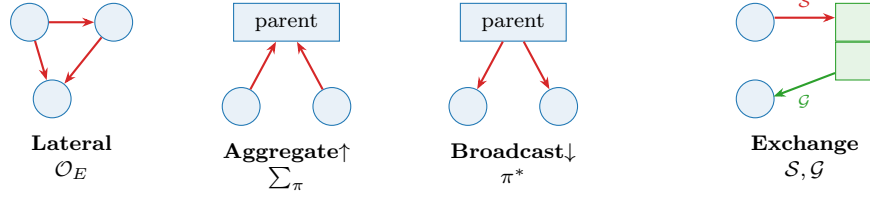


Figure 2: The four operators. *Lateral* acts within a set; *Aggregate* and *Broadcast* move across containment π ; *Exchange* scatters ($\mathcal{S}$) to and gathers ($\mathcal{G}$) from a field. P2G/G2P and the metabolite–reaction graph are both Exchange.

Composition. “MPM moves the cell” = Lateral@particle then Aggregate $\uparrow$. “Signalling moves the cell” = Lateral@cell then Broadcast $\downarrow$. “Object changes field, field moves object” = $\mathcal{S}$ then $\mathcal{G}$.

3 Dynamics: a schedule

A model is a multi-rate composition of operators, $\mathcal{O}_n \circ \dots \circ \mathcal{O}_1$, declared as a **Schedule**. Operators are pure (return deltas); the schedule sums per set and integrates, so rollout stays differentiable. The LLM agentic loop searches over schedules and operator parameterizations — one well-defined action space instead of per-domain code.

4 Glossary

Concept	Set/operator term	Symbol	Code
collection of like nodes	set (level)	S_k	<code>Level</code>
a single node	element	$x \in S_k$	row of <code>Level.state</code>
learnable per-node code	embedding	a_i	<code>Level.embedding</code>
containment	partition	$\pi_k : S_k \rightarrow S_{k+1}$	<code>Level.parent</code>
within-set links	relation	$E \subseteq S_k \times S_k$	<code>Level.edge_index</code>
continuum quantity	field	$f : \Omega \rightarrow \mathbb{R}^c$	<code>Field</code>
dynamics (GNN)	operator: ODE+Laplacian	$\mathcal{O}$	<code>Operator</code>
within-set op	lateral	$\mathcal{O}_E$	<code>Lateral</code>
up op	reduction over fibres	$\sum_{\pi}$	<code>Aggregate</code>
down op	lift	π^*	<code>Broadcast</code>
set $\leftrightarrow$ field op	transfer	$\mathcal{S}, \mathcal{G}$	<code>Exchange</code>
the container	hierarchy	$(S_{\bullet}, F_{\bullet}, \pi_{\bullet})$	<code>Hierarchy</code>
the model	composition of ops	$\mathcal{O}_n \circ \dots \circ \mathcal{O}_1$	<code>Schedule</code>